

Quasitubal SVD and Eckart–Young Optimality over Arbitrary Hilbert Spaces

A full solution of the nonseparable branch in arXiv:2504.16231

August 11, 2026

Abstract

The separability assumption in the quasitubal construction of Mor–Avron is unnecessary. For an arbitrary orthonormal-basis index set I , the correct pair is $\ell_2(I) \triangleleft \ell_\infty(I)$. We identify the quasitube algebra, construct the qSVD pointwise, and prove the exact global Eckart–Young formula for finite matrices whose entries lie in the possibly nonseparable Hilbert space. The key fact is that every individual $\ell_2(I)$ tensor has countable coordinate support.

1 Source question and answer

Section 9 of Mor–Avron asks how to “define and establish optimality results for non separable Hilbert spaces” (pages 32–33 of the arXiv PDF).

Looking forward, there are several directions to explore. From the practical perspective, there are several immediate applications of the quasitubal tensor framework that we believe are worth exploring: modeling and analyzing linear-time-invariant systems, data driven identification, control and model order reduction of continuous time dynamical systems, and so on. From a theoretical point of view, several questions remain. For example, how to define and establish optimality results for non separable Hilbert spaces, or

Banach spaces with semi-definite inner product.

We answer the nonseparable-Hilbert branch completely: replace the countable coordinate set by an arbitrary index set. The algebraic qSVD works on all coordinates, while square summability automatically makes the approximation problem countable for each concrete tensor. We treat complex Hilbert spaces; the real version is identical with orthogonal in place of unitary matrices.

2 Setup

Let $\mathcal{H}$ be an arbitrary Hilbert space, let $(e_i)_{i \in I}$ be an orthonormal basis, and let

$$F : \mathcal{H} \longrightarrow \ell_2(I), \quad (Fx)_i = \langle x, e_i \rangle$$

be the coordinate unitary. Here sums over arbitrary sets mean suprema of finite partial sums. Define

$$x \star y = F^{-1}((Fx)(Fy)), \quad x, y \in \mathcal{H},$$

where the product on the right is coordinatewise. This is well-defined because $\ell_2(I)\ell_2(I) \subset \ell_1(I) \subset \ell_2(I)$. Put $\mathcal{A} = \ell_\infty(I)$, acting diagonally on $\ell_2(I)$ and hence on $\mathcal{H}$. Thus $\mathcal{H} \cong \ell_2(I)$ is a two-sided $*$ -ideal in the commutative unital C^* -algebra $\mathcal{A}$.

For $X = (x_{ab}) \in \mathcal{A}^{m \times p}$, write

$$X(i) = (x_{ab}(i))_{a,b} \in \mathbb{C}^{m \times p}.$$

The quasitubal product and adjoint are characterized by $(XY)(i) = X(i)Y(i)$ and $X^*(i) = X(i)^*$. We use

$$\|X\|_{\text{op}} = \sup_{i \in I} \|X(i)\|_2.$$

If $X \in \mathcal{H}^{m \times p}$, its Hilbert–Frobenius norm is

$$\|X\|_{\mathcal{H},F}^2 = \sum_{a,b} \|x_{ab}\|_{\mathcal{H}}^2 = \sum_{i \in I} \|X(i)\|_F^2.$$

Finally define the implicit rank

$$\text{irank}(Y) = \sum_{i \in I} \text{rank } Y(i)$$

as a cardinal-valued sum. For finite q , $\text{irank}(Y) \leq q$ says precisely that only finitely many slices are nonzero and their matrix ranks have total at most q ; it therefore agrees with the scalar dimension of the range of the induced quasitubal multiplier in the only case used below.

3 Idea of Proof

There are three short observations. First, module endomorphisms of $\ell_2(I)$ are diagonal: the coordinate idempotents force this, and boundedness makes their diagonal an element of $\ell_\infty(I)$. Second, a qSVD is just an ordinary matrix SVD of every $X(i)$; no measurability or continuity selection is required because $\ell_\infty(I)$ contains all bounded families. Third, if the entries of X lie in $\ell_2(I)$, then $\sum_{i,j} s_j(i)^2 < \infty$. Hence the positive scalar singular values have countable support and can be sorted. Ordinary matrix Eckart–Young on every slice reduces the global problem to allocating q retained scalar singular values, whose optimum is plainly to retain the largest q .

4 Algebra and qSVD without separability

Lemma 1 (The arbitrary-index quasitube algebra). *Every $\mathcal{H}$ -module endomorphism of $(\mathcal{H}, \star)$ is automatically bounded, and these endomorphisms form a commutative unital C^* -algebra canonically and isometrically $*$ -isomorphic to $\ell_\infty(I)$. Under this identification, $a \in \ell_\infty(I)$ acts by $M_a x = F^{-1}(a F x)$ and $\|M_a\| = \|a\|_\infty$.*

Proof. Work in $\ell_2(I)$ and denote the coordinate vector by δ_i . If T is a module map, then

$$T\delta_i = T(\delta_i^2) = \delta_i T\delta_i = a_i \delta_i.$$

For arbitrary x , module linearity gives $\delta_i T x = T(\delta_i x) = a_i x_i \delta_i$; hence $T x = (a_i x_i)_i$. If (a_i) were unbounded, we could choose distinct i_n with $|a_{i_n}| \geq n^2$ and take $x = \sum_{n \geq 1} n^{-1} \delta_{i_n} \in \ell_2(I)$. Then $T x \notin \ell_2(I)$, a contradiction. Thus $a \in \ell_\infty(I)$. Conversely every bounded family a defines such a module map, and testing on δ_i gives $\|M_a\| = \sup_i |a_i|$. Composition and adjoint become coordinatewise multiplication and conjugation, proving the claim. $\square$

Lemma 2 (Arbitrary-index qSVD). *Every $X \in \mathcal{A}^{m \times p}$ has a factorization*

$$X = U S V^*,$$

where $U \in \mathcal{A}^{m \times m}$ and $V \in \mathcal{A}^{p \times p}$ are unitary and $S \in \mathcal{A}^{m \times p}$ is diagonal with $s_1(i) \geq \dots \geq s_r(i) \geq 0$ for $r = \min(m, p)$ and every $i \in I$.

Proof. For every i , choose an ordinary full matrix SVD $X(i) = U(i)S(i)V(i)^*$. Every entry of a unitary matrix has modulus at most one, and $s_j(i) \leq \|X(i)\|_2 \leq \|X\|_{\text{op}}$. Thus all chosen entries define elements of $\ell_\infty(I)$, and the pointwise identities are exactly the asserted identities over $\mathcal{A}$. $\square$

5 Global Eckart–Young theorem

Theorem 3 (Nonseparable quasitubal Eckart–Young). *Let $X \in \mathcal{H}^{m \times p}$ and let $s_j(i)$ be the singular values in an arbitrary-index qSVD. List the positive multiset*

$$\{s_j(i) : i \in I, 1 \leq j \leq r\}$$

with multiplicity in nonincreasing order $\sigma_1 \geq \sigma_2 \geq \dots > 0$, padding by zeros if the list is finite. For $q \geq 0$, let $X_{[q]}$ be obtained by retaining singular rank-one terms corresponding to the first q entries. Ties may be resolved arbitrarily, subject to retaining earlier singular directions within each slice. Then

$$\begin{aligned} \text{irank}(X_{[q]}) &\leq q, \\ \inf_{\substack{Y \in \mathcal{A}^{m \times p} \\ \text{irank}(Y) \leq q}} \|X - Y\|_{\mathcal{H}, F}^2 &= \|X - X_{[q]}\|_{\mathcal{H}, F}^2 = \sum_{n > q} \sigma_n^2, \\ \|X - X_{[q]}\|_{\text{op}} &= \sigma_{q+1} \longrightarrow 0. \end{aligned}$$

Moreover $X_{[q]}$ has entries in the span of at most q members of the chosen orthonormal basis of $\mathcal{H}$, so every truncation is finitely representable.

Proof. Unitary invariance of the Frobenius norm in every finite slice gives

$$\sum_{i \in I} \sum_{j=1}^r s_j(i)^2 = \sum_{i \in I} \|X(i)\|_F^2 = \|X\|_{\mathcal{H}, F}^2 < \infty. \quad (1)$$

An arbitrary square-summable family has countable support: for each $k \geq 1$ the set on which its absolute value is at least $1/k$ is finite, and the support is the countable union of these sets. This proves that the displayed multiset is countable. Equation (1) also implies $\sigma_n \rightarrow 0$ and permits the nonincreasing enumeration.

The construction of $X_{[q]}$ uses q scalar rank-one terms, so its implicit rank is at most q and only finitely many basis coordinates occur. Direct orthogonality yields the tail identity

$$\|X - X_{[q]}\|_{\mathcal{H}, F}^2 = \sum_{n > q} \sigma_n^2. \quad (2)$$

Now take Y with $\text{irank}(Y) \leq q$ and put $r_i = \text{rank } Y(i)$. Such a Y has finite support and belongs automatically to $\mathcal{H}^{m \times p}$. The ordinary matrix Eckart–Young theorem gives

$$\|X(i) - Y(i)\|_F^2 \geq \sum_{j > r_i} s_j(i)^2.$$

Summing in i , the right side deletes from (1) at most q members of the singular-value multiset. The largest possible deleted squared mass is $\sum_{n \leq q} \sigma_n^2$, proving from (2) that no Y improves on $X_{[q]}$.

With the stated tie convention, the spectral norm of each residual slice is its largest omitted singular value. Taking the supremum over i therefore gives $\|X - X_{[q]}\|_{\text{op}} = \sigma_{q+1}$, which tends to zero. $\square$

Corollary 4 (Separable reduction of each tensor). *For every finite $X \in \mathcal{H}^{m \times p}$ there is a separable reducing subspace $\mathcal{H}_X \subseteq \mathcal{H}$, spanned by a countable subset of the fixed basis, containing X and all its optimal finite-rank truncations. Thus the nonseparable theory is globally defined on $\ell_\infty(I)$ but every individual Hilbert-valued approximation problem is automatically separable.*

Proof. Take the union of the coordinate supports of the finitely many entries of X . Each support is countable, and all nonzero slice singular vectors and truncations remain on this union. $\square$

6 Verification, novelty, and scope

A deterministic finite-model verifier enumerates every allocation of rank budgets 0 through 8 among seven random complex matrix slices and confirms that the minimum slice-wise Eckart–Young tail equals the tail after globally sorting all scalar singular values. It also checks the coordinate ideal and diagonal multiplier inequalities. The LaTeX source, verifier output, and bounded novelty-search record accompany this packet.

The result fully answers the nonseparable-Hilbert-space branch for the orthonormal-basis tubal product used by the source. It does not address the source’s separate suggestion concerning Banach spaces with a semidefinite form, nor does it optimize the choice of orthonormal basis for a given data set.

References

- [1] U. Mor and H. Avron, arXiv:2504.16231, 2025.